



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

**School of Computer Science Engineering and
Information Systems**

M.Tech Software Engineering

SWE2008 - Android Programming

FINAL REPORT

**TITLE: HEALTHCARE APPLICATION
DEVELOPMENT**

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INTRODUCTION:

Developing healthcare applications within Android Studio demands precise development for effectiveness, user safety, and intuitive design. These apps, geared toward patient care and health information dissemination, necessitate comprehensive strategizing. This includes creating user interfaces and incorporating functionalities via Java/Kotlin. Adherence to healthcare regulations, rigorous testing, and strong security protocols are pivotal. Emphasizing privacy and smooth operations, these apps undergo continuous improvement to advance healthcare access and safeguard data. This application redefines patient-doctor interactions, fostering convenience and efficiency in healthcare management.

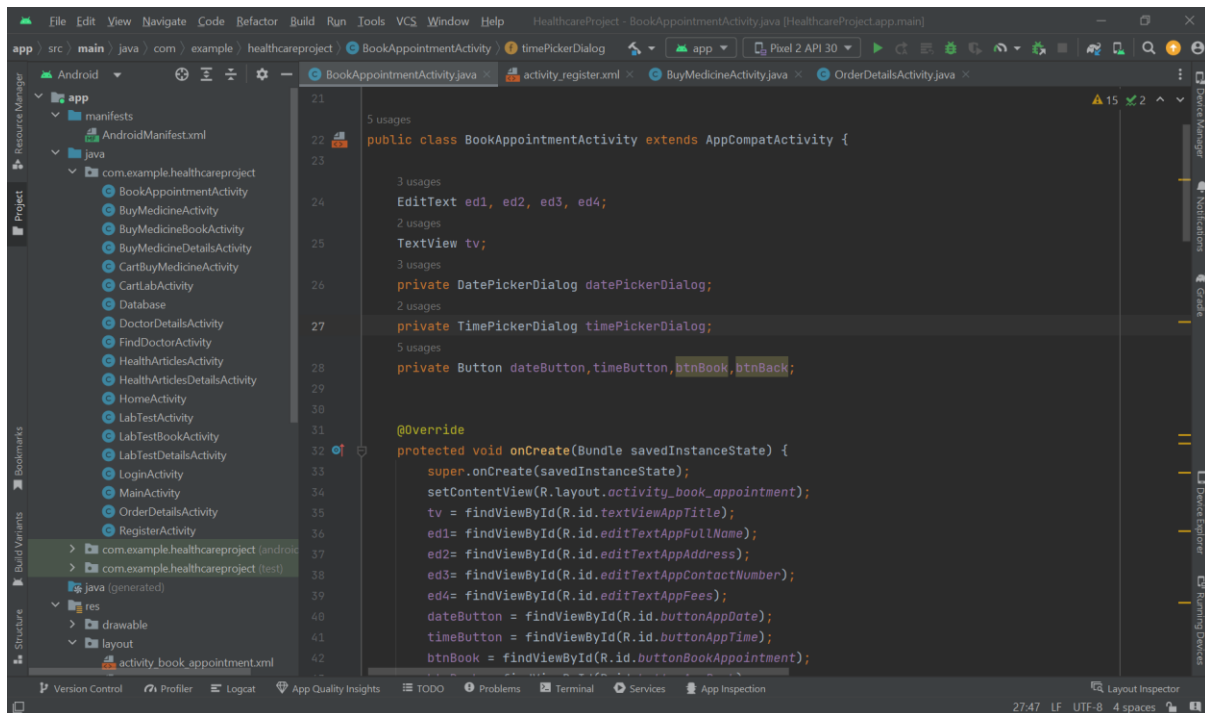
ABSTRACT:

Developing a patient booking application in Android Studio revolutionizes healthcare accessibility by facilitating easy, on-the-go appointment scheduling for patients. This app aims to streamline the healthcare process, allowing users to book appointments with healthcare providers effortlessly. Through intuitive UI/UX design and robust backend development, the app ensures a seamless experience, integrating features for appointment selection, physician availability, reminders, and possibly even telemedicine options.

ANDROID STUDIO:

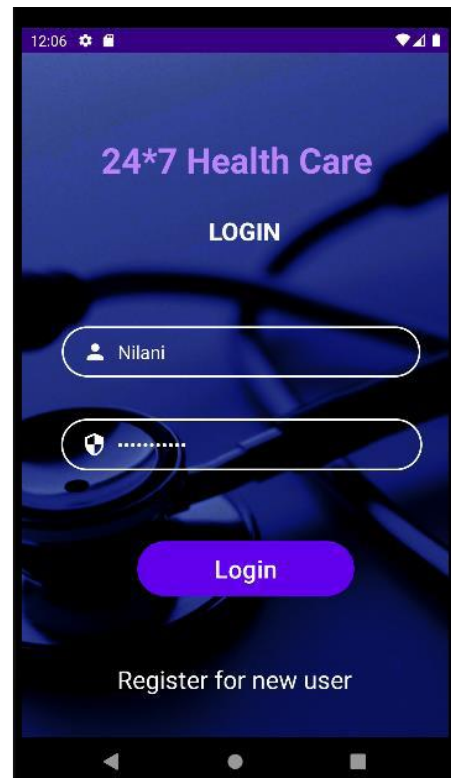
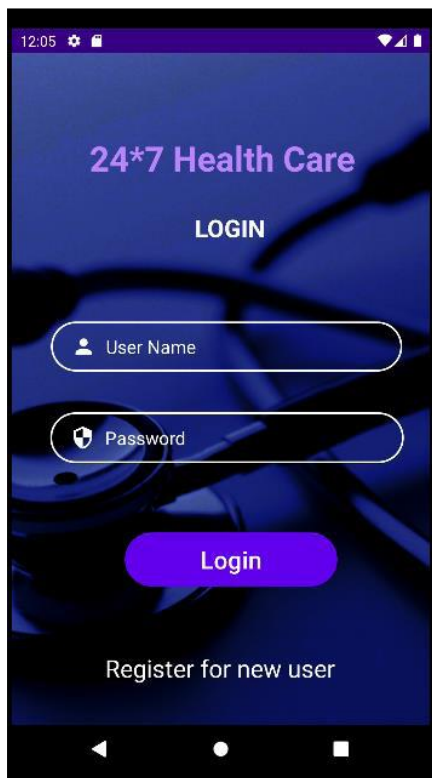
Android Studio is the official Integrated Development Environment (IDE) for Android app development, created by Google. It provides a comprehensive suite of tools for building Android applications, offering features like code editing, debugging, testing, and performance profiling. Android Studio supports various programming languages such as Java, Kotlin, and C++. It includes templates and wizards for easy project setup, an emulator for testing apps, and seamless integration with the Android SDK (Software Development Kit) for accessing APIs and tools required to develop Android applications. The IDE's user-friendly interface and continuous updates make it a preferred choice for Android developers worldwide.

ANDROID STUDIO SCREEN SHOTS:



CODE AND SCREEN SHORTS OF THE HEALTHCARE SYSTEM

LOGIN PAGE :



```

public class LoginActivity extends AppCompatActivity {

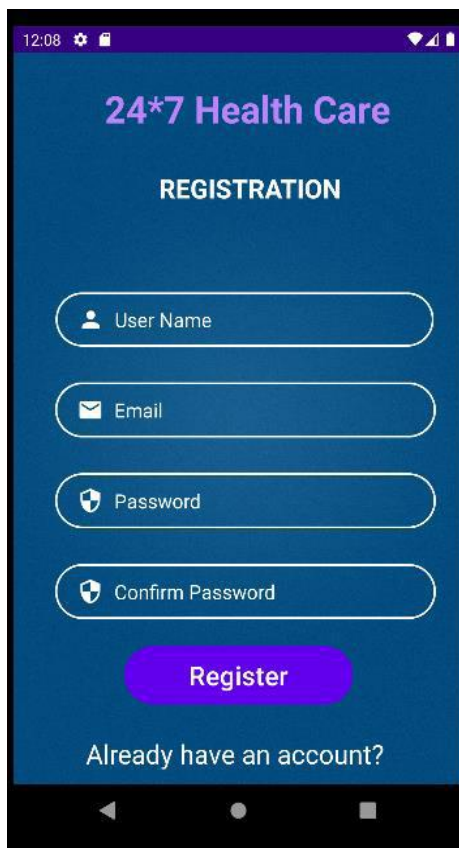
    EditText edUserName, edPassword;
    Button btn;
    TextView tv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);
        edUserName = findViewById(R.id.editTextUserName);
        edPassword = findViewById(R.id.editTextLoginPassword);
        btn = findViewById(R.id.buttonLogin);
        tv = findViewById(R.id.textViewNewUser);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                //startActivity(new Intent(LoginActivity.this,
HomeActivity.class));
                String username= edUserName.getText().toString();
                String password= edPassword.getText().toString();
                Database db = new Database(getApplicationContext(),
"healthcare", null, 1);
                if(username.length()==0 || password.length()==0)
                {
                    Toast.makeText(getApplicationContext(),"Please fill all
details", Toast.LENGTH_SHORT).show();
                }
                else {
                    if(db.login(username,password)==1){
                        Toast.makeText(getApplicationContext(),"Login
Success", Toast.LENGTH_SHORT).show();
                        SharedPreferences
sharedPreferences=getSharedPreferences("shared_prefs",
Context.MODE_PRIVATE);
                        SharedPreferences.Editor editor =
sharedPreferences.edit();
                        editor.putString("username",username);
                        editor.apply();
                        startActivity(new Intent(LoginActivity.this,
HomeActivity.class));
                    }else {
                        Toast.makeText(getApplicationContext(),"Invalid
Username and Password", Toast.LENGTH_SHORT).show();
                    }
                }
            }
        });
        tv.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(LoginActivity.this,
RegisterActivity.class));
            }
        });
    }
}

```

REGISTRATION PAGE:



The screenshot shows a mobile application interface for '24*7 Health Care'. The title '24*7 Health Care' is at the top in a light blue font. Below it, the word 'REGISTRATION' is centered in white. There are four input fields, each with a white border and a light blue background: 'User Name' (with a person icon), 'Email' (with an envelope icon), 'Password' (with a shield icon), and 'Confirm Password' (with a shield icon). Below these fields is a large, rounded, light blue button with the text 'Register' in white. At the bottom, there is a link that says 'Already have an account?' in white. The status bar at the top shows the time '12:08' and various icons. The bottom navigation bar is visible with standard Android icons.

```
public class RegisterActivity extends AppCompatActivity {

    EditText edUsername, edPassword, edEmail, edConfirm;
    Button btn;
    TextView tv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);
        edUsername = findViewById(R.id.editTextAppFullName);
        edEmail = findViewById(R.id.editTextAppAddress);
        edPassword = findViewById(R.id.editTextAppContactNumber);
        edConfirm = findViewById(R.id.editTextAppFees);
        btn = findViewById(R.id.buttonBookAppointment);
        tv = findViewById(R.id.textViewExistingUser);

        tv.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(RegisterActivity.this,
LoginActivity.class));
            }
        });

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String username = edUsername.getText().toString();
                String password = edPassword.getText().toString();
```

```

        String email = edEmail.getText().toString();
        String confirm = edConfirm.getText().toString();
        Database db = new Database(getApplicationContext(),
"healthcare", null, 1);

        if (username.length() == 0 || password.length() == 0 ||
email.length() == 0 || confirm.length() == 0) {
            Toast.makeText(getApplicationContext(), "Please fill
all details", Toast.LENGTH_SHORT).show();
        } else {
            if (password.compareTo(confirm) == 0) {
                if (isValid(password)) {
                    db.register(username, email, password);
                    Toast.makeText(getApplicationContext(), "Record
Inserted", Toast.LENGTH_SHORT).show();
                    startActivity(new Intent(RegisterActivity.this,
LoginActivity.class));
                }
            } else {
                Toast.makeText(getApplicationContext(),
"Password must contain at least 8 character having letter,digit and special
symbols.", Toast.LENGTH_SHORT).show();
            }
        }
    } else {
        Toast.makeText(getApplicationContext(), "Password
and confirm password didn't match", Toast.LENGTH_SHORT).show();
    }
}

});
}

public static boolean isValid(String passwordhere)
{
    int f1=0, f2=0, f3=0;
    if (passwordhere.length() < 8) {
        return false;
    }
    else {
        for (int p= 0; p< passwordhere.length(); p++) {
            if (Character.isLetter(passwordhere.charAt(p))) {
                f1 = 1;
            }
        }

        for (int r = 0; r < passwordhere.length(); r++) {
            if (Character.isDigit(passwordhere.charAt(r))) {
                f2 = 1;
            }
        }

        for (int s = 0; s < passwordhere.length(); s++) {
            char c = passwordhere.charAt(s);

            if (c >= 33 && c <= 46 || c == 64) {
                f3 = 1;
            }
        }
    }
}

```

```

        if (f1 == 1 && f2 == 1 && f3 == 1)
        {
            return true;
        }
        return false;
    }
}

```

HOME PAGE:



```

public class HomeActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_home);

        SharedPreferences
sharedPreferences=getSharedPreferences("shared_prefs",
Context.MODE_PRIVATE);
        String username=
sharedPreferences.getString("username","").toString();

```

```

Toast.makeText(getApplicationContext(),"Welcome"+username,Toast.LENGTH_SHORT).show();

        CardView exit= findViewById(R.id.cardExit);
        exit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                SharedPreferences.Editor editor= sharedPreferences.edit();
                editor.clear();
                editor.apply();
                startActivity(new Intent(HomeActivity.this,
LoginActivity.class));
            }
        });

        CardView findDoctor= findViewById(R.id.cardFindDoctor);
        findDoctor.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(HomeActivity.this,
FindDoctorActivity.class));
            }
        });

        CardView labTest= findViewById(R.id.cardLabTest);
        labTest.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
Intent(HomeActivity.this,LabTestActivity.class));
            }
        });

        CardView orderDetails= findViewById(R.id.cardOrderDetails);
        orderDetails.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(HomeActivity.this,
OrderDetailsActivity.class));
            }
        });

        CardView buyMedicine = findViewById(R.id.cardBuyMedicine);
        buyMedicine.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
Intent(HomeActivity.this,BuyMedicineActivity.class));
            }
        });

        CardView health = findViewById(R.id.cardHealthDoctor);
        health.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
Intent(HomeActivity.this,HealthArticlesActivity.class));
            }
        });
    }
}

```

BUY MEDICINE



```
public class BuyMedicineActivity extends AppCompatActivity {

    private String[][] packages=
    {
        {"Uprise-03 1000IU Capsule", "", "", "", "50"},
        {"HealthVit Chromium Picolinate 200mcg Capsule",
        "", "", "", "305"},
        {"Vitamin B Complex Capsules", "", "", "", "448"},
        {"Inlife Vitamin E Wheat Germ Oil Capsule", "", "", "",
        "539"},
        {"Dolo 650 Tablet", "", "", "", "30"},
        {"Crocina 650 Advance Tablet", "", "", "", "50"},
        {"Strepsils Medicated Lozenges for Sore Throat", "",
        "", "", "40"},
        {"Tata 1mg Calcium Vitamin D3", "", "", "", "30"},
        {"Feronia -XT Tablet", "", "", "", "130"},
    };

    private String[] package_details =
    {
        "Building and keeping the bones & teeth strong\n" +
        "Reducing Fatigue/stress and muscular pains\n" +
        "Boosting immunity and increasing resistance
        against infection",
        "Chromium is an essential trace mineral that
        plays an important role in helping insulin regut\n" +
        "Provides relief from vitamin B deficiencies\n"
    }
```

```
+                "Helps in formation of red blood cells\n" +  
+                "Maintains healthy nervous system",  
                "It promotes health as well as skin benefit.\n"  
  
                "It helps reduce skin blemish and  
pigmentation.\n" +  
  
                "It act as safeguard the skin from the harsh  
UVA and UVB sun rays.",  
  
                "Dolo 650 Tablet helps relieve pain and fever  
by blocking the release of certain chemical\n" +  
                "Helps relieve fever and bring down a high  
temperature\n" +  
                "Suitable for people with a heart condition or  
high blood pressure",  
  
                "Relieves the symptoms of a bacterial throat  
infection and soothes the recovery processin\n" +  
                "Provides a warm and comforting feeling during  
sore throat",  
  
                "Reduces the risk of calcium deficiency,  
Rickets, and Osteoporosis\n" +  
                "Promotes mobility and flexibility of joints",  
                "Helps to reduce the iron deficiency due to  
chronic blood loss or low intake of iron\n"  
            };  
  
    HashMap<String, String> item;  
    ArrayList list;  
    SimpleAdapter sa;  
    ListView lst;  
    Button btnBack, btnGoToCart;  
  
    @Override  
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_buy_medicine);  
  
    lst = findViewById(R.id.listViewBM);  
    btnBack= findViewById(R.id.buttonBMBack);  
    btnGoToCart=findViewById(R.id.buttonBMGoToCart);  
  
    btnGoToCart.setOnClickListener(new View.OnClickListener() {  
        @Override  
        public void onClick(View view) {  
            startActivity(new Intent(BuyMedicineActivity.this,  
CartBuyMedicineActivity.class));  
        }  
    });  
  
    btnBack.setOnClickListener(new View.OnClickListener() {  
        @Override  
        public void onClick(View view) {  
            startActivity(new Intent(BuyMedicineActivity.this,  
HomeActivity.class));  
        }  
    });  
  
    list = new ArrayList();  
for(int i=0;i< packages.length;i++){  
    item = new HashMap<String,String> ();  
    item.put("line1",packages[i][0]);
```

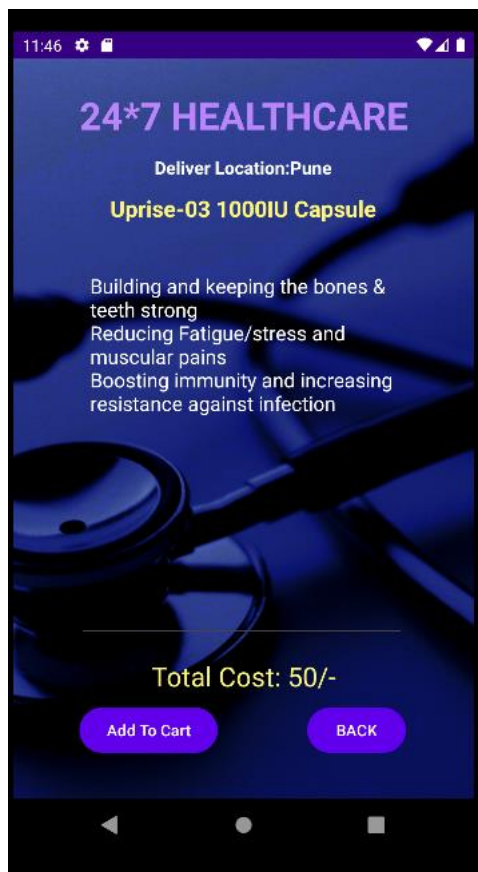
```

        item.put("line2",packages[i][1]);
        item.put("line3",packages[i][2]);
        item.put("line4",packages[i][3]);
        item.put("line5","Total Cost: "+packages[i][4]+"/-");
        list.add(item);
    }
    sa=new SimpleAdapter(this,list,
        R.layout.multi_lines,
        new String[]{"line1","line2","line3","line4","line5"},
        new
int[]{R.id.line_a,R.id.line_b,R.id.line_c,R.id.line_d,R.id.line_e,});
    lst.setAdapter(sa);

    lst.setOnItemClickListener(new AdapterView.OnItemClickListener() {
        @Override
        public void onItemClick(AdapterView<?> adapterView, View view,
int i, long l) {
            Intent it = new
Intent(BuyMedicineActivity.this,BuyMedicineDetailsActivity.class);
            it.putExtra("text1",packages[i][0]);
            it.putExtra("text2",package_details[i]);
            it.putExtra("text3",packages[i][4]);
            startActivity(it);
        }
    });
}
}

```

BUYMEDICINE<Uprise 300 1000IU CAPSULE



```

public class BuyMedicineDetailsActivity extends AppCompatActivity {

    TextView tvPackageName, tvTotalCost;
    EditText edDetails;
    Button btnBack, btnAddToCart;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_buy_medicine_details);
        tvPackageName = findViewById(R.id.textViewBMDPackageName);
        tvTotalCost = findViewById(R.id.textViewBMDTotalCost);
        edDetails = findViewById(R.id.editTextTextBMDMultiLine);
        btnAddToCart = findViewById(R.id.buttonBMDAddToCart);
        btnBack = findViewById(R.id.buttonBMDBack);
        edDetails.setKeyListener(null);

        Intent intent = getIntent();
        tvPackageName.setText(intent.getStringExtra("text1"));
        edDetails.setText(intent.getStringExtra("text2"));
        tvTotalCost.setText("Total Cost:
"+intent.getStringExtra("text3")+"/-");

        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
Intent(BuyMedicineDetailsActivity.this, BuyMedicineActivity.class));
            }
        });

        btnAddToCart.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                SharedPreferences sharedPreferences =
getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
                String username = sharedPreferences.getString("username",
"" ).toString();
                String product = tvPackageName.getText().toString();
                float price =
Float.parseFloat(intent.getStringExtra("text3").toString());

                Database db = new Database(getApplicationContext(),
"healthcare", null, 1);

                if (db.checkCart(username, product) == 1) {
                    Toast.makeText(getApplicationContext(), "Product
already added", Toast.LENGTH_SHORT).show();
                } else {
                    db.addCart(username, product, price, "medicine");
                    Toast.makeText(getApplicationContext(), "Record
inserted to cart", Toast.LENGTH_SHORT).show();
                    startActivity(new
Intent(BuyMedicineDetailsActivity.this, BuyMedicineActivity.class));
                }
            }
        });
    }
}

```

BUYMEDICINE<GOTOCART<CHECKOUT

```
public class BuyMedicineBookActivity extends AppCompatActivity {

    EditText edname, edaddress, edcontact, edpincode;
    Button btnBooking;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_buy_medicine_book);

        edname=findViewById(R.id.editTextBMBFullname);
        edaddress=findViewById(R.id.editTextBMBAAddress);
        edcontact=findViewById(R.id.editTextBMBCContact);
        edpincode=findViewById(R.id.editTextBMBCPincode);
        btnBooking=findViewById(R.id.buttonBMBCBooking);

        Intent intent=getIntent();
        String[] price =
intent.getStringExtra("price").toString().split(java.util.regex.Pattern.quote(":"));
        String date = intent.getStringExtra("date");
        //String time = intent.getStringExtra("time");

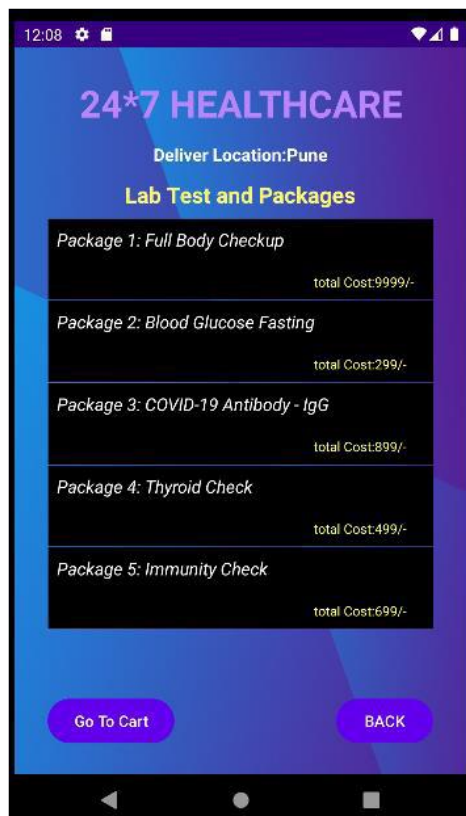
        btnBooking.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                SharedPreferences sharedPreferences=
getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
                String username =
sharedPreferences.getString("username", "").toString();
```

```

        Database db= new
Database(getApplicationContext(),"healthcare",null,1);
        db.addOrder(username,
edname.getText().toString(),edaddress.getText().toString(),edcontact.getTex
t().toString(),Integer.parseInt(edpincode.getText().toString()),date.toStri
ng()," ",Float.parseFloat(price[1].toString()),"medicine");
        db.removeCart(username,"medicine");
        Toast.makeText(getApplicationContext(),"Your Booking Done
Successfully",Toast.LENGTH_SHORT).show();
        startActivity(new Intent(BuyMedicineBookActivity.this,
HomeActivity.class));
    }
    });
}
}

```

LABTEST:



```

public class LabTestActivity extends AppCompatActivity {
    private String[][] packages=
    {
        {"Package 1: Full Body Checkup", "", "", "", "9999"},
        {"Package 2: Blood Glucose Fasting", "", "", "", "299"},
        {"Package 3: COVID-19 Antibody - IgG", "", "", "", "899"},
        {"Package 4: Thyroid Check", "", "", "", "499"},
        {"Package 5: Immunity Check", "", "", "", "699"}
    };
    private String[] package_details= {
        "Blood Glucose Fasting\n"
        + "Complete Hemogram" +
    }
}

```

```

        "HbA1c\n"
        + "Iron Studies\n"
        + "Kidney Function Test\n"
        + "LOW Lactate Dehydrogenase, Serum\n"
        + "Lipid Profile\n"
        + "Liver Function test", "Blood Glucose Fasting",
"COVTD-19 Antibody- IgG",
        "Thyroid Profile-Total (T3, T4 & TSH Ultra-sensitive)",
        "Complete Hemogram\n"
        + "CRP (C Reactive Protein) Quantitative, Serum\n"
        + "Iron Studies\n"
        + "Kidney Function Test\n"
        + "Vitamin D Total-25 Hydroxyin\n"
        + "Liver Function Test\n"
        + "Lipid Profile"
    };

    HashMap<String,String> item;
    ArrayList list;
    SimpleAdapter sa;
    Button btnGoToCart,btnBack;
    ListView listView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_lab_test);
        btnGoToCart = findViewById(R.id.buttonLTGoToCart);
        btnBack = findViewById(R.id.buttonLTBack);
        listView = findViewById(R.id.listViewLT);
        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
Intent(LabTestActivity.this,HomeActivity.class));
            }
        });

        list= new ArrayList<>();
        for (int i=0;i<packages.length;i++){
            item=new HashMap<String,String>();
            item.put("line1",packages[i][0]);
            item.put("line2",packages[i][1]);
            item.put("line3",packages[i][2]);
            item.put("line4",packages[i][3]);
            item.put("line5","total Cost:"+packages[i][4]+"/-");
            list.add(item);
        }
        sa = new SimpleAdapter(this,list, R.layout.multi_lines,new
String[]{"line1","line2","line3","line4","line5"},
            new int
[]{R.id.line_a,R.id.line_b,R.id.line_c,R.id.line_d,R.id.line_e});
        listView.setAdapter(sa);

        listView.setOnItemClickListener(new
AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> adapterView, View view,
int i, long l) {
                Intent it = new

```

```

Intent (LabTestActivity.this, LabTestDetailsActivity.class);
        it.putExtra("text1", packages[i][0]);
        it.putExtra("text2", package_details[i]);
        it.putExtra("text3", packages[i][4]);
        startActivity(it);
    }
});

    btnGoToCart.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            startActivity(new Intent (LabTestActivity.this,
CartLabActivity.class));
        }
    });
}
}

```

LABTEST< PACKAGE1: FULL BODY CHECK UP



```

public class LabTestDetailsActivity extends AppCompatActivity {

    TextView tvPackageName,tvTotalCost;
    EditText edDetails;
    Button btnAddToCart,btnBack;

```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_lab_test_details);
    tvPackageName = findViewById(R.id.textViewLDPackageName);
    tvTotalCost = findViewById(R.id.textViewLDTotalCost);
    edDetails = findViewById(R.id.editTextLDTextMultiLine);
    btnAddToCart = findViewById(R.id.buttonLDAddToCart);
    btnBack = findViewById(R.id.buttonLDBack);

    edDetails.setKeyListener(null);

    Intent intent = getIntent();
    tvPackageName.setText(intent.getStringExtra("text1"));
    edDetails.setText(intent.getStringExtra("text2"));
    tvTotalCost.setText("Total Cost:
    "+intent.getStringExtra("text3")+"/-");
    btnBack.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            startActivity(new
    Intent(LabTestDetailsActivity.this, LabTestActivity.class));
        }
    });
    btnAddToCart.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            SharedPreferences sharedPreferences =
    getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
            String username = sharedPreferences.getString("username",
    "").toString();
            String product = tvPackageName.getText().toString();
            float price =
    Float.parseFloat(intent.getStringExtra("text3").toString());

            Database db = new Database(getApplicationContext(),
    "healthcare", null, 1);

            if (db.checkCart(username, product) == 1) {
                Toast.makeText(getApplicationContext(), "Product
    already added", Toast.LENGTH_SHORT).show();
            } else {
                db.addCart(username, product, price, "lab");
                Toast.makeText(getApplicationContext(), "Record
    inserted to cart", Toast.LENGTH_SHORT).show();
                startActivity(new Intent(LabTestDetailsActivity.this,
    LabTestActivity.class));
            }
        }
    });
}
}

```

LABTEST<GOTOCART<CHECKOUT

24*7 Health Care

Please Fill All Details For Booking

Full Name

Address

Pincode

Contact Number

BOOK

```
public class LabTestBookActivity extends AppCompatActivity {

    EditText edname, edaddress, edcontact, edpincode;
    Button btnBooking;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_lab_test_book);

        edname=findViewById(R.id.editTextTBFullname);
        edaddress=findViewById(R.id.editTextTBAddress);
        edcontact=findViewById(R.id.editTextTBContact);
        edpincode=findViewById(R.id.editTextTBPincode);
        btnBooking=findViewById(R.id.buttonLTBBooking);

        Intent intent=getIntent();
        String[] price =
intent.getStringExtra("price").toString().split(java.util.regex.Pattern.quote(":"));
        String date = intent.getStringExtra("date");
        String time = intent.getStringExtra("time");

        btnBooking.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                SharedPreferences sharedPreferences=
getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
                String username =
```

```

sharedPreferences.getString("username","").toString();
        Database db= new
Database(getApplicationContext(),"healthcare",null,1);
        db.addOrder(username,
edname.getText().toString(),edaddress.getText().toString(),edcontact.getTex
t().toString(),Integer.parseInt(edpincode.getText().toString()),date.toStri
ng(),time.toString(),Float.parseFloat(price[1].toString()),"lab");
        db.removeCart(username,"lab");
        Toast.makeText(getApplicationContext(),"Your Booking Done
Successfully",Toast.LENGTH_SHORT).show();
        startActivity(new Intent(LabTestBookActivity.this,
HomeActivity.class));
    }
    });
}
}

```

FIND DOCTOR



```

public class FindDoctorActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_find_doctor);

        CardView exit= findViewById(R.id.cardFDBack);
        exit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(FindDoctorActivity.this,
HomeActivity.class));
            }
        });
    }
}

```

```

        CardView familyphysician= findViewById(R.id.cardFDFamilyPhysician);
        familyphysician.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(FindDoctorActivity.this,
DoctorDetailsActivity.class);
                it.putExtra("title", "Family Physician");
                startActivity(it);
            }
        });

        CardView dietician = findViewById(R.id.cardFDDietician);
        dietician.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(FindDoctorActivity.this,
DoctorDetailsActivity.class);
                it.putExtra("title", "Dietician");
                startActivity(it);
            }
        });

        CardView dentist= findViewById(R.id.cardFDDentist);
        dentist.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(FindDoctorActivity.this,
DoctorDetailsActivity.class);
                it.putExtra("title", "Dentist");
                startActivity(it);
            }
        });

        CardView surgeon= findViewById(R.id.cardFDSurgeon);
        surgeon.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(FindDoctorActivity.this,
DoctorDetailsActivity.class);
                it.putExtra("title", "Surgeon");
                startActivity(it);
            }
        });

        CardView cardiologist = findViewById(R.id.cardFDCardiologist);
        cardiologist.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(FindDoctorActivity.this,
DoctorDetailsActivity.class);
                it.putExtra("title", "Cardiologist");
                startActivity(it);
            }
        });
    }
}

```

FIND DOCTOR<FAMILY PHYSICIAN



```
public class DoctorDetailsActivity extends AppCompatActivity {
    TextView tv;
    Button btn;
    private String[][] doctor_details1=
    {
        {"Doctor Name: Ajit Saste", "Hospital Address: Pimpri",
"Exp: 5yrs", "Mobile No:9898989898", "600"},
        {"Doctor Name: Prasad Pawar", "Hospital Address:
Nigdi", "Exp: 15yrs", "Mobile No: 7898989898", "900"},
        {"Doctor Name: Swapnil Kale", "Hospital Address: Pune",
"Exp: 8yrs", "Mobile No:8898989898", "300"},
        {"Doctor Name: Deepak Deshmukh", "Hospital Address:
Chinchwad", "Exp: 6yrs", "Mobile No:9898000000", "500"},
        {"Doctor Name: Ashok Panda", "Hospital Address:
Katrai", "Exp: 7yrs", "Mobile No:7798989898", "800"}
    };
    private String[][] doctor_details2=
    {
        {"Doctor Name: Ajit Saste", "Hospital Address: Pimpri",
"Exp: 5yrs", "Mobile No:9898989898", "600"},
        {"Doctor Name: Prasad Pawar", "Hospital Address:
Nigdi", "Exp: 15yrs", "Mobile No: 7898989898", "900"},
        {"Doctor Name: Swapnil Kale", "Hospital Address: Pune",
"Exp: 8yrs", "Mobile No:8898989898", "300"},
        {"Doctor Name: Deepak Deshmukh", "Hospital Address:
Chinchwad", "Exp: 6yrs", "Mobile No:9898000000", "500"},
        {"Doctor Name: Ashok Panda", "Hospital Address:
Katrai", "Exp: 7yrs", "Mobile No:7798989898", "800"}
    };
}
```

```

private String[][] doctor_details3=
{
    {"Doctor Name: Ajit Saste", "Hospital Address: Pimpri",
"Exp: 5yrs", "Mobile No:9898989898", "600"},
    {"Doctor Name: Prasad Pawar", "Hospital Address:
Nigdi", "Exp: 15yrs", "Mobile No: 7898989898", "900"},
    {"Doctor Name: Swapnil Kale", "Hospital Address: Pune",
"Exp: 8yrs", "Mobile No:8898989898", "300"},
    {"Doctor Name: Deepak Deshmukh", "Hospital Address:
Chinchwad", "Exp: 6yrs", "Mobile No:9898000000", "500"},
    {"Doctor Name: Ashok Panda", "Hospital Address:
Katrai", "Exp: 7yrs", "Mobile No:7798989898", "800"}
};

private String[][] doctor_details4=
{
    {"Doctor Name: Ajit Saste", "Hospital Address: Pimpri",
"Exp: 5yrs", "Mobile No:9898989898", "600"},
    {"Doctor Name: Prasad Pawar", "Hospital Address:
Nigdi", "Exp: 15yrs", "Mobile No: 7898989898", "900"},
    {"Doctor Name: Swapnil Kale", "Hospital Address: Pune",
"Exp: 8yrs", "Mobile No:8898989898", "300"},
    {"Doctor Name: Deepak Deshmukh", "Hospital Address:
Chinchwad", "Exp: 6yrs", "Mobile No:9898000000", "500"},
    {"Doctor Name: Ashok Panda", "Hospital Address:
Katrai", "Exp: 7yrs", "Mobile No:7798989898", "800"}
};

private String[][] doctor_details5=
{
    {"Doctor Name: Ajit Saste", "Hospital Address: Pimpri",
"Exp: 5yrs", "Mobile No:9898989898", "600"},
    {"Doctor Name: Prasad Pawar", "Hospital Address:
Nigdi", "Exp: 15yrs", "Mobile No: 7898989898", "900"},
    {"Doctor Name: Swapnil Kale", "Hospital Address: Pune",
"Exp: 8yrs", "Mobile No:8898989898", "300"},
    {"Doctor Name: Deepak Deshmukh", "Hospital Address:
Chinchwad", "Exp: 6yrs", "Mobile No:9898000000", "500"},
    {"Doctor Name: Ashok Panda", "Hospital Address:
Katrai", "Exp: 7yrs", "Mobile No:7798989898", "800"}
};

String[][] doctor_details = {};
ArrayList list;
HashMap<String, String> item;
SimpleAdapter sa;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_doctor_details);

    tv= findViewById(R.id.textView_logo6);
    btn = findViewById(R.id.buttonLDBack);

    Intent it= getIntent();
    String title= it.getStringExtra("title");
    tv.setText(title);

    if(title.compareTo("Family Physician")==0)
        doctor_details= doctor_details1;
    else
    if(title.compareTo("Dietician")==0)
        doctor_details= doctor_details2;

```

```

else
if(title.compareTo("Dentist")==0)
    doctor_details= doctor_details3;
else
if(title.compareTo("Surgeon")==0)
    doctor_details= doctor_details4;
else
doctor_details=doctor_details5;

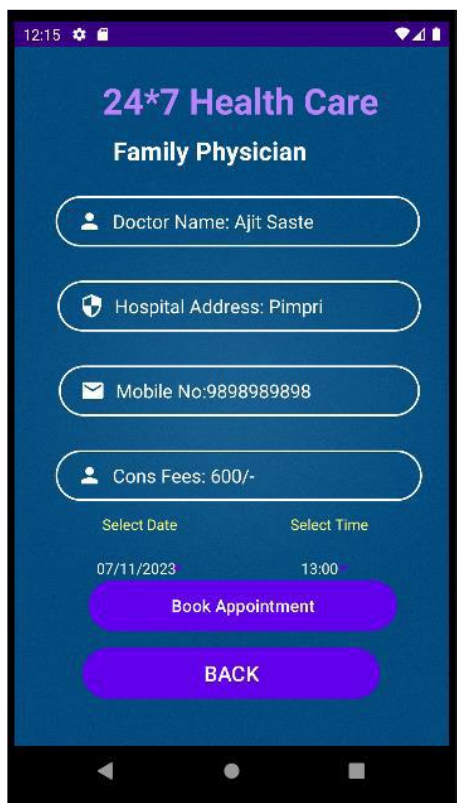
btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        startActivity(new Intent(DoctorDetailsActivity.this,
FindDoctorActivity.class));
    }
});

list= new ArrayList<>();
for(int i=0; i<doctor_details.length;i++)
{
    item= new HashMap<String, String>();
    item.put("line1",doctor_details[i][0]);
    item.put("line2",doctor_details[i][1]);
    item.put("line3",doctor_details[i][2]);
    item.put("line4",doctor_details[i][3]);
    item.put("line5","Cons Fee: "+doctor_details[i][4]+"/-");
    list.add(item);
}
sa = new SimpleAdapter(this,list, R.layout.multi_lines,new
String[]{"line1","line2","line3","line4","line5"},
    new int
[] {R.id.line_a,R.id.line_b,R.id.line_c,R.id.line_d,R.id.line_e});

ListView lst = findViewById(R.id.listViewLT);
lst.setAdapter(sa);
lst.setOnItemClickListener(new AdapterView.OnItemClickListener() {
    @Override
    public void onItemClick(AdapterView<?> adapterView, View view,
int i, long l) {
        Intent it = new Intent(DoctorDetailsActivity.this,
BookAppointmentActivity.class);
        it.putExtra("text1", title);
        it.putExtra("text2", doctor_details[i][0]);
        it.putExtra("text3", doctor_details[i][1]);
        it.putExtra("text4", doctor_details[i][3]);
        it.putExtra("text5", doctor_details[i][4]);
        startActivity(it);
    }
});
}
}

```

FIND DOCTOR> FAMILY PHYSICIAN> BOOK APPOINTMENT



```
public class BookAppointmentActivity extends AppCompatActivity {

    EditText ed1, ed2, ed3, ed4;
    TextView tv;
    private DatePickerDialog datePickerDialog;
    private TimePickerDialog timePickerDialog;
    private Button dateButton, timeButton, btnBook, btnBack;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_book_appointment);
        tv = findViewById(R.id.textViewAppTitle);
        ed1= findViewById(R.id.editTextAppFullName);
        ed2= findViewById(R.id.editTextAppAddress);
        ed3= findViewById(R.id.editTextAppContactNumber);
        ed4= findViewById(R.id.editTextAppFees);
        dateButton = findViewById(R.id.buttonAppDate);
        timeButton = findViewById(R.id.buttonAppTime);
        btnBook = findViewById(R.id.buttonBookAppointment);
        btnBack = findViewById(R.id.buttonAppBack);

        ed1.setKeyListener(null);
        ed2.setKeyListener(null);
        ed3.setKeyListener(null);
        ed4.setKeyListener(null);

        Intent it = getIntent();
        String title = it.getStringExtra("text1");
        String fullname = it.getStringExtra("text2");
```

```

String address = it.getStringExtra("text3");
String contact= it.getStringExtra("text4");
String fees = it.getStringExtra("text5");

tv.setText(title);
ed1.setText(fullname);
ed2.setText(address);
ed3.setText(contact);
ed4.setText("Cons Fees: "+fees+"/-");

//datePicker
initDatePicker();

dateButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        datePickerDialog.show();
    }
});

//timePicker
initTimePicker();

timeButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        timePickerDialog.show();
    }
});

btnBack.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        startActivity(new
Intent(BookAppointmentActivity.this, FindDoctorActivity.class));
    }
});

btnBook.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {

        Database db = new Database(getApplicationContext(),
"healthcare", null, 1);
        SharedPreferences sharedPreferences =
getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
        String username = sharedPreferences.getString("username",
 "").toString();

        if (db.checkAppointmentExists(username, title+" =>
"+fullname, address, contact, dateButton.getText().toString(), timeButton.getTe
xt().toString()) == 1) {

            Toast.makeText(getApplicationContext(), "Appointment
Already Booked", Toast.LENGTH_SHORT).show();
        } else {
            db.addOrder(username, title+" =>
"+fullname, address, contact, 0, dateButton.getText().toString(), timeButton.get
Text().toString(), Float.parseFloat(fees), "appointment");
            Toast.makeText(getApplicationContext(), "Your
Appointment is done Successfully", Toast.LENGTH_SHORT).show();
            startActivity(new Intent(BookAppointmentActivity.this,

```

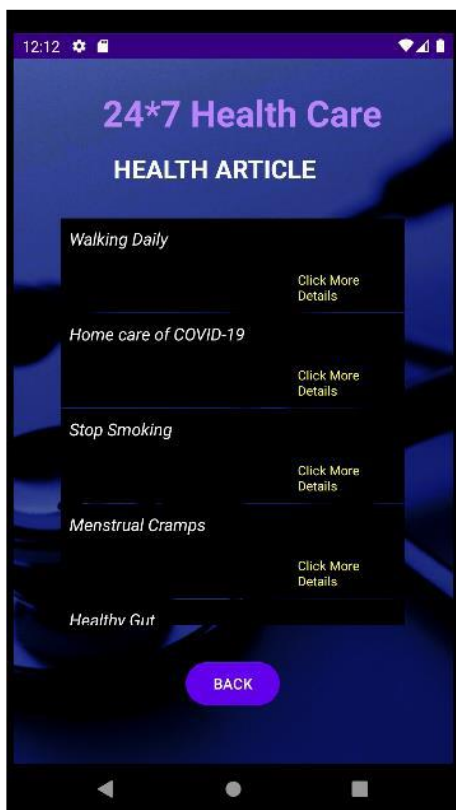
```

HomeActivity.class));
        }
    }
});
}
private void initDatePicker(){
    DatePickerDialog.OnDateSetListener dateSetListener = new
DatePickerDialog.OnDateSetListener() {
    @Override
    public void onDateSet(DatePicker datePicker, int i, int i1, int i2)
{
        i1 = i1 + 1;
        dateButton.setText(i2+"/"+i1+"/"+i);
    }
};
Calendar cal = Calendar.getInstance();
int year= cal.get(Calendar.YEAR);
int month= cal.get(Calendar.MONTH);
int day = cal.get(Calendar.DAY_OF_MONTH);
int style= AlertDialog.THEME_HOLO_DARK;
datePickerDialog = new
DatePickerDialog(this,style,dateSetListener,year,month,day);

datePickerDialog.getDatePicker().setMinDate(cal.getTimeInMillis()+86400000)
;
}
private void initTimePicker(){
    TimePickerDialog.OnTimeSetListener timeSetListener= new
TimePickerDialog.OnTimeSetListener() {
    @Override
    public void onTimeSet(TimePicker timePicker, int i, int i1) {
        timeButton.setText(i+":"+i1);
    }
};
Calendar cal = Calendar.getInstance();
int hrs = cal.get(Calendar.HOUR);
int mins= cal.get(Calendar.MINUTE);
int style= AlertDialog.THEME_HOLO_DARK;
timePickerDialog= new
TimePickerDialog(this,style,timeSetListener,hrs,mins,true);
}
}

```

HEALTH ARTICLE:



```
public class HealthArticlesActivity extends AppCompatActivity {

    private String[][] health_details = {
        {"Walking Daily", "", "", "", "Click More Details"},
        {"Home care of COVID-19", "", "", "", "Click More Details"},
        {"Stop Smoking", "", "", "", "Click More Details"},
        {"Menstrual Cramps", "", "", "", "Click More Details"},
        {"Healthy Gut", "", "", "", "Click More Details"},
    };

    private int[] images= {

        R.drawable.health1,
        R.drawable.health2,
        R.drawable.health3,
        R.drawable.health4,
        R.drawable.health5
    };

    HashMap<String,String> item;
    ArrayList list;
    SimpleAdapter sa;
    Button btnBack;
    ListView lst;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_health_article);
    }
}
```

```

        lst = findViewById(R.id.listViewHA);
        btnBack= findViewById(R.id.buttonHABack);

        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
Intent(HealthArticlesActivity.this,HomeActivity.class));
            }
        });

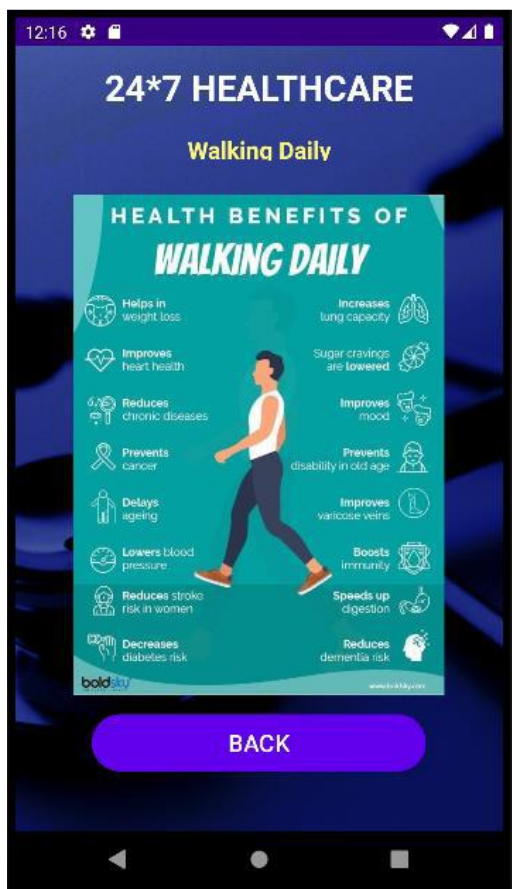
        list = new ArrayList();
        for(int i=0;i< health_details.length;i++){
            item = new HashMap<String,String>();
            item.put("line1",health_details[i][0]);
            item.put("line2",health_details[i][1]);
            item.put("line3",health_details[i][2]);
            item.put("line4",health_details[i][3]);
            item.put("line5",health_details[i][4]);
            list.add(item);
        }

        sa=new SimpleAdapter(this,list,
            R.layout.multi_lines,
            new String[]{"line1","line2","line3","line4","line5"},
            new
int[]{R.id.line_a,R.id.line_b,R.id.line_c,R.id.line_d,R.id.line_e,});
        lst.setAdapter(sa);

        lst.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> adapterView, View view,
int i, long l) {
                Intent it = new
Intent(HealthArticlesActivity.this,HealthArticlesDetailsActivity.class);
                it.putExtra("text1",health_details[i][0]);
                it.putExtra("text2",images[i]);
                startActivity(it);
            }
        });
    }
}

```

HEALTH ARTICLES> WALKING DAILY



```
public class HealthArticlesDetailsActivity extends AppCompatActivity {

    TextView tv1;
    ImageView img;
    Button btnBack;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_health_articles_details);

        btnBack= findViewById(R.id.buttonHADBack);
        tv1 =findViewById(R.id.textViewHADTitle);
        img = findViewById(R.id.imageViewHAD);

        Intent intent=getIntent();
        tv1.setText(intent.getStringExtra("text1"));
        Bundle bundle= getIntent().getExtras();
        if(bundle != null)
        {
            int resId = bundle.getInt("text2");
            img.setImageResource(resId);
        }
        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new
                Intent(HealthArticlesDetailsActivity.this, HealthArticlesActivity.class));
            }
        });
    }
}
```

```
}  
}
```

ORDER DETAILS



```
public class OrderDetailsActivity extends AppCompatActivity {  
  
    private String[][] order_details = {};  
    HashMap<String, String> item;  
    ArrayList list;  
    SimpleAdapter sa;  
    ListView lst;  
    Button btn;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_order_details);  
        btn= findViewById(R.id.buttonODBack);  
        lst = findViewById(R.id.listViewOD);  
  
        btn.setOnClickListener(new View.OnClickListener() {  
            @Override  
            public void onClick(View view) {  
                startActivity(new Intent(OrderDetailsActivity.this,  
HomeActivity.class));  
            }  
        });  
  
        Database db= new  
Database(getApplicationContext(),"healthcare",null,1);  
        SharedPreferences sharedPreferences=
```

```

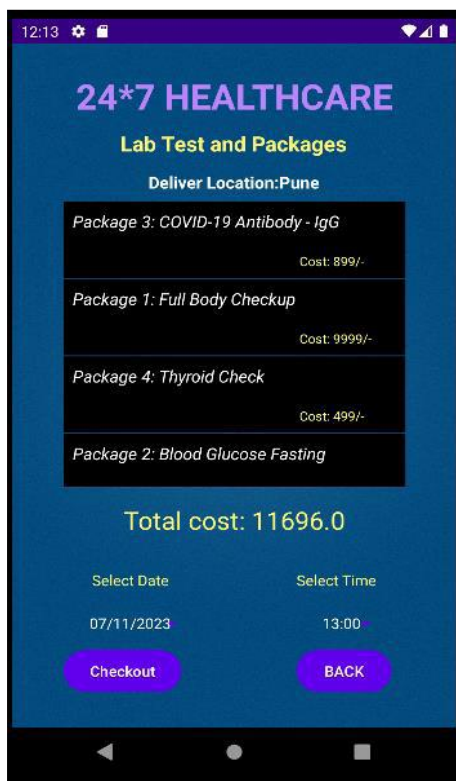
getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
    String username =
sharedPreferences.getString("username","").toString();
    ArrayList dbData= db.getOrderData(username);

    order_details = new String[dbData.size()][5];
    for(int i=0;i<order_details.length;i++) {
        order_details[i] = new String[5];
        String arrData = dbData.get(i).toString();
        String[] strData =
arrData.split(java.util.regex.Pattern.quote("$"));
        order_details[i][0] = strData[0];
        order_details[i][1] = strData[1];//+" " + strData[3];
        if (strData[7].compareTo("medicine") == 0) {
            order_details[i][3] = "Del:" + strData[4];

        } else {
            order_details[i][3] = "Del:" + strData[4] + " " +
strData[5];
        }
        order_details[i][2] = "Rs." + strData[6];
        order_details[i][4] = strData[7];
    }
    list = new ArrayList();
    for(int i=0;i<order_details.length;i++){
        item = new HashMap<String,String>();
        item.put("line1",order_details[i][0]);
        item.put("line2",order_details[i][1]);
        item.put("line3",order_details[i][2]);
        item.put("line4",order_details[i][3]);
        item.put("line5",order_details[i][4]);
        list.add(item);
    }
    sa=new SimpleAdapter(this,list,
        R.layout.multi_lines,
        new String[]{"line1","line2","line3","line4","line5"},
        new
int[]{R.id.line_a,R.id.line_b,R.id.line_c,R.id.line_d,R.id.line_e,});
    lst.setAdapter(sa);
}
}

```

CART



```
public class CartBuyMedicineActivity extends AppCompatActivity {

    HashMap<String , String> item;
    ArrayList list;
    SimpleAdapter sa;
    TextView tvTotal;
    ListView lst;
    private DatePickerDialog datePickerDialog;
    private TimePickerDialog timePickerDialog;
    private Button dateButton, btnCheckout, btnBack;
    private String[][] packages={};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_cart_buy_medicine);

        dateButton= findViewById(R.id.buttonBMCartDate);

        btnCheckout= findViewById(R.id.buttonBMCartCheckout);
        btnBack= findViewById(R.id.buttonBMCartBack);
        tvTotal=findViewById(R.id.textViewBMCartTotalCost);
        lst = findViewById(R.id.listViewBMCart);

        SharedPreferences sharedPreferences=
        getSharedPreferences("shared_prefs", Context.MODE_PRIVATE);
        String username = sharedPreferences.getString("username",
        "").toString();
        Database db= new
        Database(getApplicationContext(),"healthcare",null,1);
        float totalAmount = 0;
```

```

        ArrayList dbData= db.getCartData(username,"medicine");

//Toast.makeText(getApplicationContext(),""+dbData,Toast.LENGTH_SHORT).show
();
        packages= new String[dbData.size()][5];
        for(int i=0; i<packages.length; i++)
        {
            packages[i]= new String[5];
        }
        for(int i=0;i<dbData.size();i++)
        {
            String arrData = dbData.get(i).toString();
            String[] strData =
arrData.split(java.util.regex.Pattern.quote("$"));
            packages[i][0]= strData[0];
            packages[i][4]= "Cost: "+strData[1]+"/-";
            totalAmount= totalAmount+Float.parseFloat(strData[1]);
        }

        tvTotal.setText("Total cost: "+totalAmount);

        list=new ArrayList<>();
        for(int i=0; i<packages.length;i++)
        {
            item = new HashMap<String, String>();
            item.put("line1",packages[i][0]);
            item.put("line2",packages[i][1]);
            item.put("line3",packages[i][2]);
            item.put("line4",packages[i][3]);
            item.put("line5",packages[i][4]);
            list.add(item);
        }

        sa=new SimpleAdapter(this, list,
            R.layout.multi_lines,
            new String[]{"line1","line2","line3","line4","line5"},
            new
int[]{R.id.line_a,R.id.line_b,R.id.line_c,R.id.line_d,R.id.line_e,});

        lst.setAdapter(sa);

        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(CartBuyMedicineActivity.this,
BuyMedicineActivity.class));
            }
        });

        btnCheckout.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(CartBuyMedicineActivity.this,
BuyMedicineBookActivity.class);
                it.putExtra("price",tvTotal.getText());
                it.putExtra("data",dateButton.getText());
                startActivity(it);
            }
        });
    }
}

```

```

        initDatePicker();

        dateButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                datePickerDialog.show();
            }
        });
    }

    private void initDatePicker(){
        DatePickerDialog.OnDateSetListener dateSetListener = new
        DatePickerDialog.OnDateSetListener() {
            @Override
            public void onDateSet(DatePicker datePicker, int i, int i1, int
i2) {
                i1 = i1 + 1;
                dateButton.setText(i2+"/"+i1+"/"+i);
            }
        };
        Calendar cal = Calendar.getInstance();
        int year= cal.get(Calendar.YEAR);
        int month= cal.get(Calendar.MONTH);
        int day = cal.get(Calendar.DAY_OF_MONTH);
        int style= AlertDialog.THEME_HOLO_DARK;
        datePickerDialog = new
        DatePickerDialog(this, style, dateSetListener, year, month, day);

        datePickerDialog.getDatePicker().setMinDate(cal.getTimeInMillis()+86400000)
        ;
    }
}

```

WIDGETS USED:

List View:

ListView is a UI component used to display a scrollable list of items. It's commonly used to present a vertically scrollable collection of data, allowing users to view and interact with a set of items, such as text, images, or custom layouts.

The ListView is populated using an Adapter, which acts as a bridge between the underlying data source (such as an array or a database) and the ListView itself. The adapter manages the content of the list and determines how each item in the list is displayed.

Card View:

CardView is a UI component that helps in presenting information or content in a structured and visually appealing format resembling cards. It's used to encapsulate content within a rounded-corner rectangle, creating a card-like

appearance, and is often employed to display elements such as images, text, buttons, or other widgets.

The CardView is part of the Android Material Design library and offers features like shadow effects and corner radius, enhancing the visual hierarchy and making the content stand out against the background. Developers can customize the elevation (shadow depth) and corner radius of the CardView based on the design requirements.

Button:

Button is a clickable user interface element used to trigger actions within an app. Customizable in appearance, it allows users to interact by tapping, executing specific tasks on press.

Text Field:

TextField or EditText is a user interface element that enables users to input and edit text. It provides a space for users to enter information, such as text, numbers, or passwords, within an app's interface. Developers can customize its appearance and behavior, specifying attributes like text size, color, input type, and validation rules. Users can interact with the TextField by typing, editing, or deleting text as needed.

Linear layout:

LinearLayout is a view group that arranges its child views in a single direction—either horizontally or vertically—based on the orientation set. It's used to create a linear arrangement of UI components, stacking them one after the other in a sequence. This layout allows for simple organization and positioning of elements in a linear fashion within an app's interface. Developers can specify the arrangement (vertical or horizontal) and define the layout parameters for each child view to control their positioning and size within the linear layout.

Date picker:

DatePicker is a user interface component that enables users to select a date from a graphical calendar interface. It presents a widget where users can choose the year, month, and day, allowing them to input or modify dates conveniently. Developers can integrate this component into their apps to facilitate date selection for various purposes such as setting appointments, scheduling tasks, or capturing specific dates within the application.

Time picker:

TimePicker is a user interface element that allows users to select a specific time using a graphical interface. It presents a widget with hour and minute controls, enabling users to set or modify the time conveniently. Developers often incorporate the TimePicker into their apps for tasks like setting reminders, scheduling events, or capturing time-related data within the application.

SQLite Database:

SQLite is a lightweight, embedded relational database management system used in Android development and various other software applications. It's a self-contained, serverless, and transactional SQL database engine that is embedded directly into the application.

In Android development, SQLite is the primary database used for storing and managing structured data within the app. It's a part of the Android framework and is widely used due to its simplicity, efficiency, and compatibility with Android devices.

Key features of SQLite in Android development include:

- **Embedded Database:** SQLite operates as a local database within the app, stored in a single file directly on the device. This simplifies setup and allows the app to manage its own data without needing a separate database server.
- **SQL Compatibility:** SQLite supports SQL (Structured Query Language) for creating, querying, updating, and managing data. It allows developers to perform various operations like creating tables, inserting, updating, deleting, and querying data using SQL commands.
- **Efficiency:** It's designed to be lightweight and efficient, consuming minimal system resources and requiring minimal setup and administration.
- **Transactional Support:** SQLite supports ACID (Atomicity, Consistency, Isolation, Durability) properties, ensuring reliability and data integrity through transactional operations.

Developers use SQLite in Android apps to store structured data such as user information, settings, application data, and more. It provides a simple yet powerful way to manage local data storage, enabling efficient data handling within Android applications.

CODE FOR THE DATABASE CREATED FOR HEALTHCARE PROJECT:

```
public class Database extends SQLiteOpenHelper {
    public Database(@Nullable Context context, @Nullable String name,
@Nullable SQLiteDatabase.CursorFactory factory, int version) {
        super(context, name, factory, version);
    }

    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        String qry1="create table users(username text, email text, password
text)";
        sqLiteDatabase.execSQL(qry1);
        String qry2="create table cart(username text, product text, price
float, otype text)";
        sqLiteDatabase.execSQL(qry2);
        String qry3="create table orderplace(username text, fullname text,
address text, contactnumber text, pincode int, date text, time text, amount
float, otype text)";
        sqLiteDatabase.execSQL(qry3);
    }

    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int i1) {
    }
    public void register(String username, String email, String password){
        ContentValues cv= new ContentValues();
        cv.put("username", username);
        cv.put("email", email);
        cv.put("password", password);
        SQLiteDatabase db = getWritableDatabase();
        db.insert("users", null, cv);
        db.close();
    }
    public int login(String username, String password){
        int result = 0;
        String str[]= new String[2];
        str[0]=username;
        str[1]=password;
        SQLiteDatabase db= getReadableDatabase();
        Cursor c = db.rawQuery("select * from users where username=? and
password=?",str);
        if(c.moveToFirst()){
            result=1;
        }

        return result;
    }
    public void addCart(String username, String product, float price,
String otype){
        ContentValues cv = new ContentValues();
        cv.put("username",username);
        cv.put("product",product);
        cv.put("price",price);
        cv.put("otype",otype);
        SQLiteDatabase db = getWritableDatabase();
    }
```

```

        db.insert("cart",null,cv);
        db.close();
    }

    public int checkCart(String username,String product){
        int result = 0;
        String str[] = new String[2];
        str[0] = username;
        str[1] = product;
        SQLiteDatabase db = getReadableDatabase();
        Cursor c = db.rawQuery("select * from cart where username = ? and
product = ?",str);
        if (c.moveToFirst()){
            result = 1;
        }
        db.close();
        return result;
    }

    public void removeCart(String username,String otype){
        String str[]= new String[2];
        str[0]= username;
        str[1]= otype;
        SQLiteDatabase db = getWritableDatabase();
        db.delete("cart","username=? and otype=?",str);
        db.close();
    }

    public ArrayList getCartData(String username, String otype){
        ArrayList<String> arr= new ArrayList<>();
        SQLiteDatabase db= getReadableDatabase();
        String str[]= new String[2];
        str[0]=username;
        str[1]= otype;
        Cursor c = db.rawQuery("select * from cart where username = ? and
otype = ?",str);
        if(c.moveToFirst())
        {
            do{
                String product = c.getString(1);
                String price = c.getString(2);
                arr.add(product+"$"+ price);
            }
            while (c.moveToNext());
        }
        db.close();
        return arr;
    }

    public void addOrder(String username, String fullname, String address,
String contact, int pincode, String date, String time,float amount, String
otype)
    {
        ContentValues cv = new ContentValues();
        cv.put("username", username);
        cv.put("fullname", fullname);
        cv.put("address", address);
        cv.put("contactno", contact);
        cv.put("pincode", pincode);
        cv.put("date", date);
    }

```

```

        cv.put("time", time);
        cv.put("amount", amount);
        cv.put("otype", otype);
        SQLiteDatabase db = getWritableDatabase();
        db.insert("orderplace", null, cv);
        db.close();
    }

    public ArrayList getOrderData(String username) {
        ArrayList<String> arr= new ArrayList<>();
        SQLiteDatabase db = getReadableDatabase();
        String str[]= new String[1];
        str[0]=username;
        Cursor c = db.rawQuery("select * from orderplace where username = ?",str);
        if(c.moveToFirst()){
            do{
                arr.add(c.getString(1)+"$"+c.getString(2)+"$"+c.getString(3)+"$"+c.getString(4)+"$"+c.getString(5)+"$"+c.getString(6)+"$"+c.getString(7)+"$"+c.getString(8));
            } while (c.moveToFirst());
        }

        db.close();
        return arr;
    }

    public int checkAppointmentExists(String username, String fullname, String address, String contact, String date, String time){

        int result = 0;
        String str[]=new String[6];
        str[0]=username;
        str[1]=fullname;
        str[2]=address;
        str[3]=contact;
        str[4]=date;
        str[5]=time;
        SQLiteDatabase db = getReadableDatabase();
        Cursor c = db.rawQuery("select * from orderplace where username = ? and fullname = ? and address = ? and contactno = ? and date = ? and time = ?",str);
        if(c.moveToFirst())
        {
            result=1;
        }
        db.close();
        return result;
    }
}

```